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Synopsis
SmartPharm

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering
In
Artificial Intelligence and Machine Learning

by

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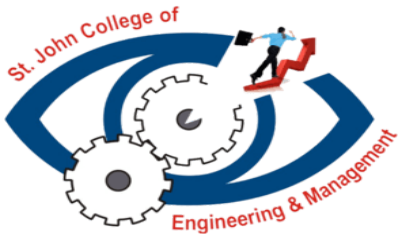
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Under the Guidance of

_____

_____



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI
2026-2027

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This is to certify that Neev Borana [EU1246063], Romit Meher [EU1226023] and Rohit Gorhekar [EU1246019] are bonafide students of the Department of Artificial Intelligence and Machine Learning, St. John College of Engineering and Management. They have submitted the project SmartPharm for evaluation during the academic year 2026-2027.

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Principal,
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Acknowledgement

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Group Members

1. Neev Borana
2. Romit Meher
3. Rohit Gorhekar

ABSTRACT

SmartPharm is a web-based pharmacy management project that connects medicine stocking, batch inventory, POS billing, procurement, prescription review and operational insight. The design separates a medicine identity from its manufactured batch and from the physical quantity stored at a location.

The implementation uses a Next.js / React / TypeScript frontend, Flask and SQLAlchemy APIs, Aiven MySQL for business records and Aiven PostgreSQL for AI/reference data. Barcode input, camera capture and packaging OCR provide working review-assisted identification. Prescription and invoice reading produce editable drafts for verification.

Billing validates stock and expiry, allocates eligible batches in FEFO order and commits stock, sale and payment records together. Dashboard views show recent invoices, stock and expiry alerts, purchase drafts and replenishment estimates. PWA support provides installation controls and a safe reconnect experience.

The classroom demonstration contains eight medicines and sixteen paired package photos, with labelled synthetic transactions. Verification includes focused automated tests and deployed browser checks. Recognition output remains subject to human review; no untested accuracy percentage or autonomous clinical capability is claimed.

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Abbreviations and symbols

- **ML**: Machine Learning
- **AI**: Artificial Intelligence
- **TF-IDF**: Term Frequency-Inverse Document Frequency
- **NLP**: Natural Language Processing
- **DFD**: Data Flow Diagram
- **UML**: Unified Modeling Language
- **CSV**: Comma-Separated Values
- **UI**: User Interface
- **PDF**: Portable Document Format
- **HR**: Human Resources
- **ETL**: Extract, Transform, Load

CHAPTER 1: INTRODUCTION

1.1 Background

SmartPharm is a pharmacy management web application that connects medicine stocking, batch-level inventory, billing, procurement, prescription review and operational insight. A medicine identity is separated from its manufactured batches and from physical quantity at a storage location. This distinction supports changing batch prices and expiry dates.

1.2 Problem Statement

The project addresses repeated manual entry and disconnected medicine, inventory and billing records.

- Existing problem: staff must identify the intended medicine while tracking strength, batch, expiry and physical availability.
- Industrial issue: product-level stock totals do not explain differences between batch prices, expiry dates or shelf locations.
- Research gap: OCR output alone does not establish a saleable medicine identity or a valid stock transaction.
- Need for improvement: connect reviewed identification with catalogue matching, FEFO billing and traceable stock changes.

1.3 Need of the Project

The project supports consistent pharmacy records and reduces repeated typing through barcode input, camera capture and document reading. Human review and manual search keep uncertain recognition recoverable.

1.4 Objectives

Develop an integrated pharmacy workflow that remains usable when optional recognition or network services cannot provide a reliable result.
Maintain separate medicine, batch and location-stock records with store-scoped access.
Provide barcode lookup and working review-assisted photo recognition for medicine entry and billing.

- Apply FEFO allocation and validate stock and expiry before accepting a sale.
- Support reviewed procurement, eligible returns and audited stock adjustments.
- Present recent activity, expiry alerts and explainable replenishment estimates.
- Verify stock/payment invariants, offline replay behaviour and responsive user interfaces.

1.5 Scope

The project covers counter and back-office workflows for a pharmacy chain installation.

- Included: catalogue, inventory, POS, purchases, suppliers, prescriptions, returns, shelf locations, staff access, expenses and dashboard.
- Boundary: recognition assists review; it does not independently prescribe or make final dispensing decisions.
- Future expansion includes independently evaluated image recognition, physical-device testing, backup restoration exercises and longer-history forecast validation.

1.6 Expected Outcomes

The project provides a deployed web prototype with connected business records and a repeatable classroom demonstration.

- Working prototype: responsive browser interface and PWA installation support.
- Software: Next.js / React frontend, Flask APIs, Aiven MySQL business records and Aiven PostgreSQL AI references.
- Demonstration: eight medicines represented by sixteen paired package photos.
- Traceability: medicine, batch, location, sale and purchase records remain distinct.
- Operational output: recent invoice chart, low-stock and expiry alerts, purchase drafts and reorder estimates.
- Verification output: scoped automated tests and deployed browser checks.
- Academic contribution: integration of reviewed recognition with transactional pharmacy operations.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Seven primary research papers are surveyed on OCR, scene-text detection, handwriting, document layout and retrieval-augmented generation. Findings describe the cited research rather than measured SmartPharm recognition accuracy.

2.2 Literature Review Table

The literature comparison follows the prescribed paper, author, year, method, findings and research-gap structure.

Paper	Author	Year	Method	Findings	Research Gap
[1] CRNN	Shi, Bai and Yao	2015	CNN + recurrent sequence recognition	End-to-end image text sequence modelling	Text must still be linked to a valid medicine identity.
[2] Differentiable Binarization	Liao et al.	2019 / 2020	Learned text-region binarisation	Efficient scene text detection	Detection is not product identification.
[3] PP-OCR	Du et al.	2020	Lightweight detector, direction classifier, recogniser	Practical OCR pipeline	Foil, rotation and glare need application handling.
[4] TrOCR	Li et al.	2021	Pretrained image and text Transformers	Printed and handwritten transcription	Ambiguous names/directions require review.
[5] LayoutLMv3	Huang et al.	2022	Unified text/image masking and alignment	Document layout and content representation	Project uses OCR layout parsing, not this whole model.
[6] PaddleOCR 3.0	Cui et al.	2025	OCR, structured parsing and extraction	Maintained document processing toolkit	Toolkit benchmarks do not establish project accuracy.
[7] RAG	Lewis et al.	2020	Generator with retrieved external memory	Grounded knowledge-intensive generation	Retrieved content still needs validation and provenance.

The selected papers combine foundational recognition research with recent OCR and document-understanding work.

2.3 Research Gap

Previous research provides strong methods for recognising text and representing document content.

- CRNN and Transformer recognisers turn image regions into text; detection methods locate candidate text regions.
- Unresolved application concerns include reflective medicine foil, rotated labels, ambiguous handwriting and linking extracted text to verified store stock.
- SmartPharm addresses the integration gap with catalogue evidence, manual recovery, review before confirmation and batch-aware transactions.

2.4 Novelty / Innovation

The engineering contribution is a connected review workflow rather than a new OCR architecture.

- Medicine entry and billing share reviewed identity mappings and barcode aliases.
- Purchase drafts are separate from accepted physical intake; FEFO billing uses recorded batch expiry.
- A low-evidence scan leads to recapture or manual search rather than an invented medicine identity.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System

A Next.js and React frontend communicates with Flask APIs over HTTPS. SQLAlchemy services validate and persist business operations in Aiven MySQL. Aiven PostgreSQL supports AI reference/vector retrieval. Optional Google Gemini and Tavily integrations assist handwriting and reference search.

3.2 Working Principle

1. Sign in and retrieve the current store and role. 2. Enter a medicine manually or read a barcode/package. 3. Review identity and batch details before stock entry. 4. Select a stocked medicine and quantity in POS. 5. Validate saleable batches and allocate FEFO stock. 6. Commit sale, tender records and stock deduction together. 7. Review dashboard alerts, purchase drafts and eligible returns.

3.3 System Architecture / Block Diagram

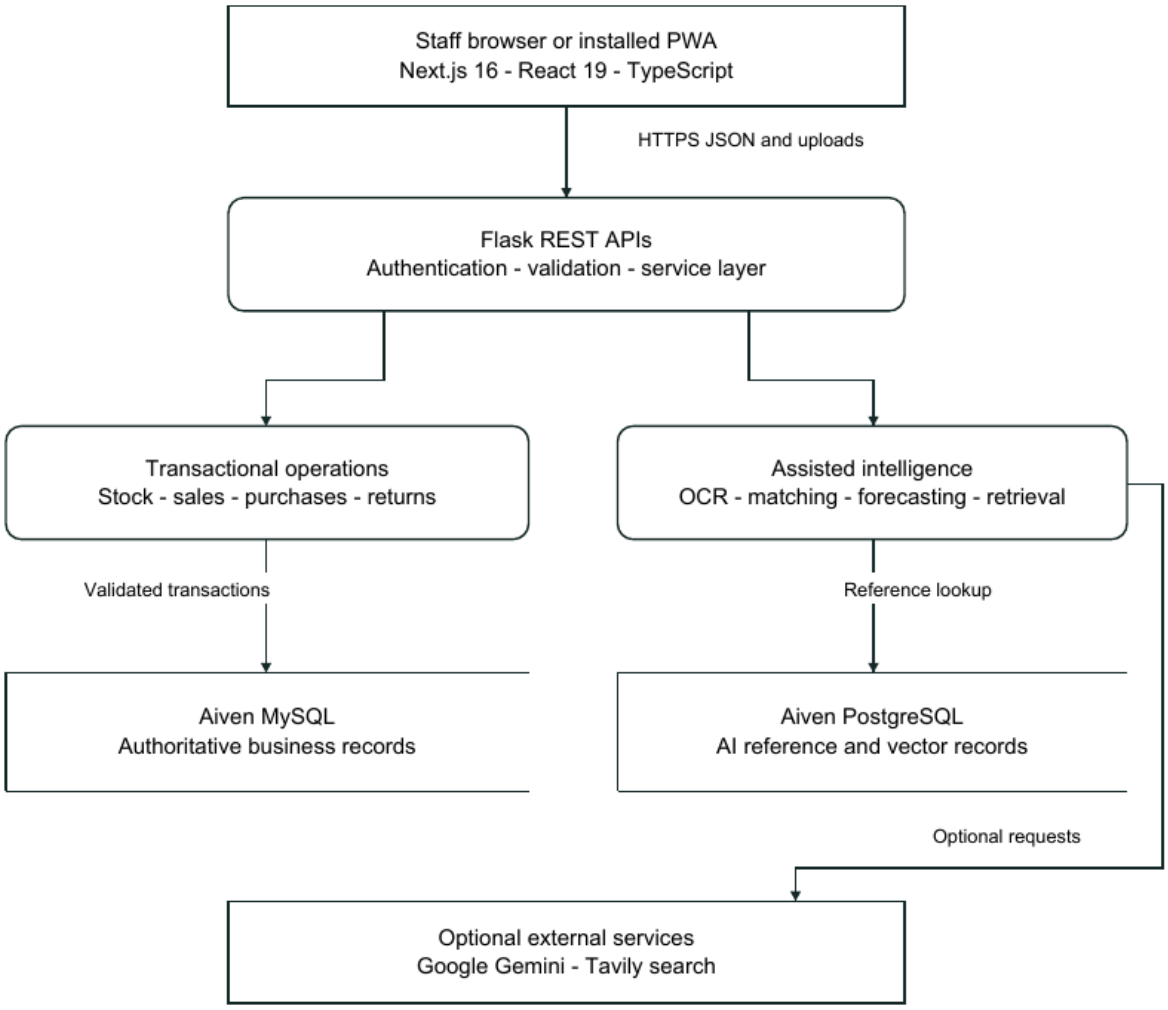


Figure 1 System Architecture

3.4 Flow Chart

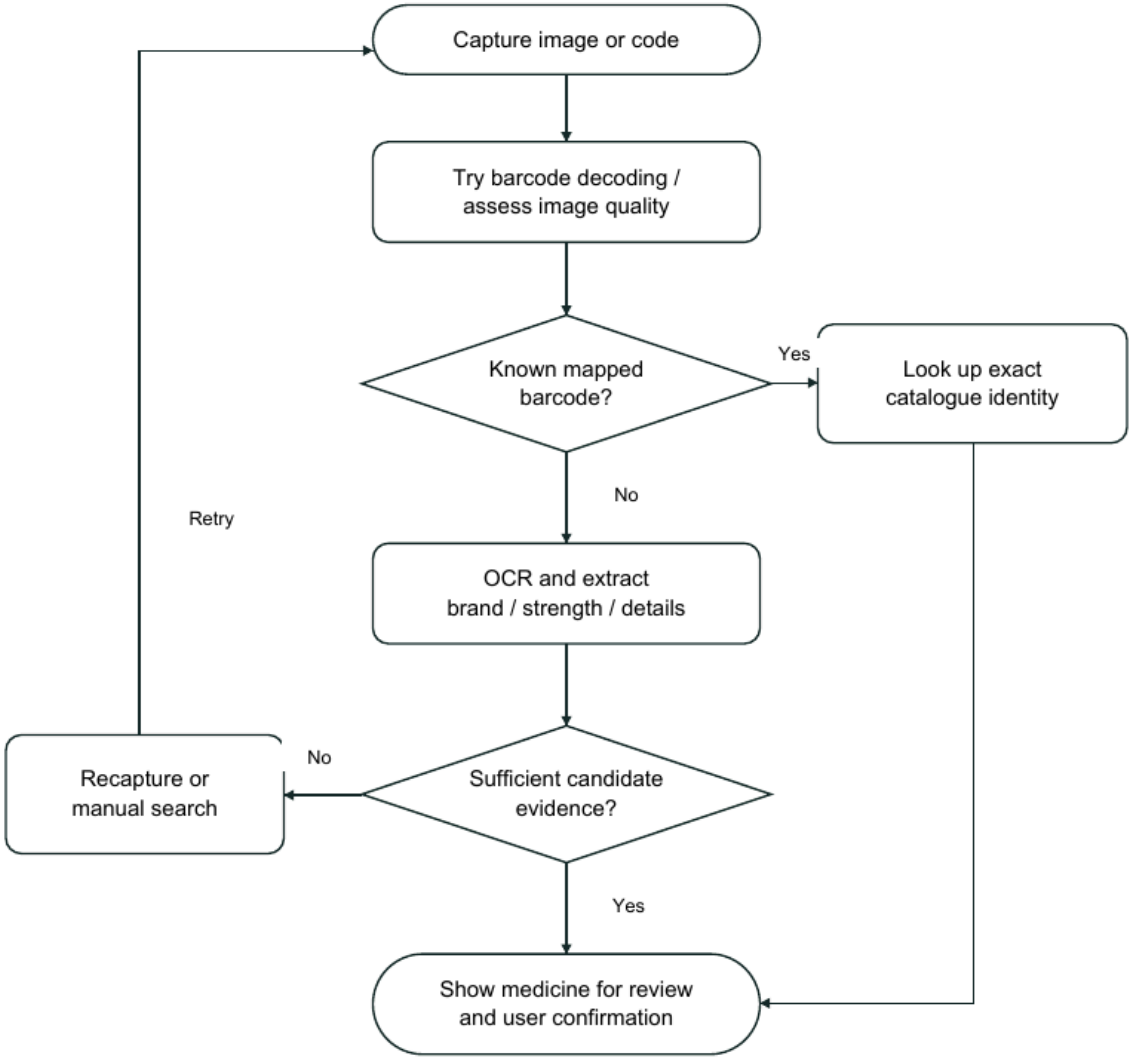


Figure 2 Recognition Flowchart

Implementation uses incremental service, API, UI and regression-test changes.
The design preserves authoritative business data and separates optional recognition results from confirmed stock.

- Design procedure: model medicine identity, batch pricing/expiry, location quantities and transaction links.
- Experimental procedure: prepare isolated fixtures, exercise deterministic operations, inspect deployed read-only paths and review scan outcomes.
- Algorithms: catalogue evidence matching, barcode alias lookup, FEFO allocation, adaptive moving-average demand estimation and Ridge regression when sufficient selling-day history exists.
- Physical workflow: compare actual delivered quantities with invoice drafts before accepting stock.
- Programming steps: implement validation and service transactions; expose authenticated routes; connect review forms; add focused tests; build and verify deployment.

3.6 Components / Materials

The principal implementation components are listed below.

Component	Inspected implementation or requirement
Frontend	Next.js 16.3.5, React 19.2.8, TypeScript 5 and responsive CSS.

Backend	Python Flask, SQLAlchemy, Alembic migrations and a production WSGI server.
Vision / ML	PaddleOCR, OpenCV, NumPy, scikit-learn; optional experimental PyTorch training.
Databases	Aiven MySQL for business data; Aiven PostgreSQL for AI references/vector data.
Hosting	Separate frontend/API containers on Google Cloud Run over HTTPS.
Capture / devices	Desktop or phone browser with camera permission; optional USB keyboard-style barcode scanner.
Development GPU	RTX 3070 available for local experiments; not a requirement for Cloud Run CPU inference.

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure

Work packages comprise requirements/schema, transactional services, responsive UI, camera/document input, billing and returns, planning views, deployment, tests and academic documentation.

4.2 Gantt Chart

Suggested semester milestones are shown below; this is a planning schedule, not a retrospectively measured development log.

Work package	W1-2	W3-4	W5-6	W7-8	W9-10	W11-12
Requirements and data design	X	X				
Core stock and APIs		X	X			
Billing and procurement UI			X	X		
OCR and review flows				X	X	
Deployment and verification					X	X
Report and demonstration						X

4.3 Roles and Responsibilities

The group members participate in project review and demonstration. Individual development contributions should be confirmed with the project guide.

Student	PID	Responsibility
Neev Borana	EU1246063	Group review and demonstration; individual allocation to be confirmed.
Romit Meher	EU1226023	Group review and demonstration; individual allocation to be confirmed.
Rohit Gorhekar	EU1246019	Group review and demonstration; individual allocation to be confirmed.

4.4 Risk Assessment

Risk	Impact	Mitigation
Ambiguous label or handwriting	Wrong candidate	Reject weak matches; retain original image and manual review.
Interrupted network / repeated request	Duplicate or pending bill	Persistent UUID queue and server replay receipt.
Thin sales history	Unstable estimate	Baseline fallback and user-reviewed draft orders.
Credential exposure	Unauthorised access	Keep secrets outside code; rotate exposed credentials.

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

Established web, database and OCR libraries make the prototype technically feasible. Manual search remains available when recognition is uncertain. MySQL is used for business persistence and PostgreSQL for AI/reference data. Production services are containerised on Google Cloud Run.

5.2 Economic Feasibility

Existing computers and a camera can be reused. Costs depend on hosting, database plans, image volume, API quotas, backups and printing. Free tiers can support a classroom demo but are not permanent cost guarantees.

5.3 Operational Feasibility

Staff must understand batch entry, quantity units, expiry eligibility and reconciliation. The interface supports phone/desktop layouts and review before state changes. Physical-device and operational-load evaluation remain separate from isolated tests.

5.4 Budget Estimation

Item	Quantity	Cost basis
Existing development computer	As available	Existing resource; invoice not supplied.
Camera / scanner	Camera; optional scanner	Reuse available equipment; quote if purchased.
Cloud and database services	Usage / plan based	Monitor actual billing and quota.
API services	Optional usage	Free-tier eligibility and paid rates can change.
Printing and binding	2 group copies	Obtain local print-shop quotation.

5.5 Bill of Materials (BOM) (if any(

No dedicated manufactured hardware assembly is required. A computer or supported phone browser, network connection and camera are sufficient for the core demonstration; a keyboard-style barcode scanner is optional.

CHAPTER 6: EXPECTED RESULTS

The expected outcomes below are supported by implemented review flows and the scoped verification evidence in the appendix.
The project presents recognition as a working review-assisted feature, with a recovery path for uncertain input.

- Prototype: connected catalogue, stocking, FEFO billing, purchases, returns and operational dashboard.
- Accuracy: exact mapped barcode lookup and evidence-based candidate review; independent image-recognition accuracy is not expressed as an untested percentage.
- Efficiency: reduce repeated entry through camera, barcode, catalogue search and editable invoice/prescription drafts.
- Performance: use bounded public-asset PWA caching, fresh protected requests and a simpler forecast baseline for short history.
- Simulation: labelled synthetic sales, suppliers, customers, purchase drafts and expenses populate the demonstration.
- Verification: focused tests check balanced stock/payment import, idempotency, sparse forecasts, camera lifecycle and browser layout.

Detailed test cases, recorded outcomes, API integrations and actual frontend screenshots are included in the appendices.

CHAPTER 7: SUSTAINABILITY & IMPACT

7.1 Industrial Relevance

Batch-level stock, reviewed procurement, expiry handling and consistent transaction history are relevant to physical pharmacy operations.

7.2 Societal Impact

Structured records and pharmacist review can support safer data entry. The project does not replace professional clinical judgement or establish measured patient outcomes.

7.3 Environmental Impact

Expiry visibility and replenishment planning may help reduce avoidable disposal. Digital records can reduce routine paper use. No carbon or waste-reduction measurement is claimed.

7.4 SDG Mapping

The following SDG connections are qualitative project mappings.
SDG 3: organised medicine records and pharmacist verification.
SDG 9: connected web and data infrastructure for operations.
SDG 12: expiry review and demand-informed stock planning.
Operational impact depends on accurate data and staff procedures.
Cloud/model resource use should be monitored alongside potential paper savings.
Sustainability outcomes require measurement during real operational use.

CHAPTER 8: CONCLUSION

SmartPharm connects pharmacy records through reviewed identification and batch-aware operations.

- The problem is inconsistent identity, expiry and quantity information across disconnected workflows.
- The solution integrates barcode/photo assistance, a store catalogue, FEFO billing, reviewed intake and traceable transactions.
- The contribution is a deployed academic prototype with repeatable demo records and scoped verification. Future work includes recognition evaluation across capture conditions, longer-history forecasts, concurrent-terminal testing and backup restoration.

REFERENCES

The following references use numbered citations corresponding to the literature survey.
Research papers

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- [5] Y. Huang, T. Lv, L. Cui, Y. Lu and F. Wei, "LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking," ACM Multimedia, 2022, doi:10.1145/3503161.3548112. https://arxiv.org/abs/2204.08387

[6] C. Cui et al., "PaddleOCR 3.0 Technical Report," arXiv:2507.05595, 2025. https://arxiv.org/abs/2507.05595
[7] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS, 2020. https://arxiv.org/abs/2005.11401

APPENDICES (Optional)

- The following appendices contain additional labelled diagrams, API integrations, verification cases and eight actual frontend screens.
- Additional diagrams

Activity Diagram

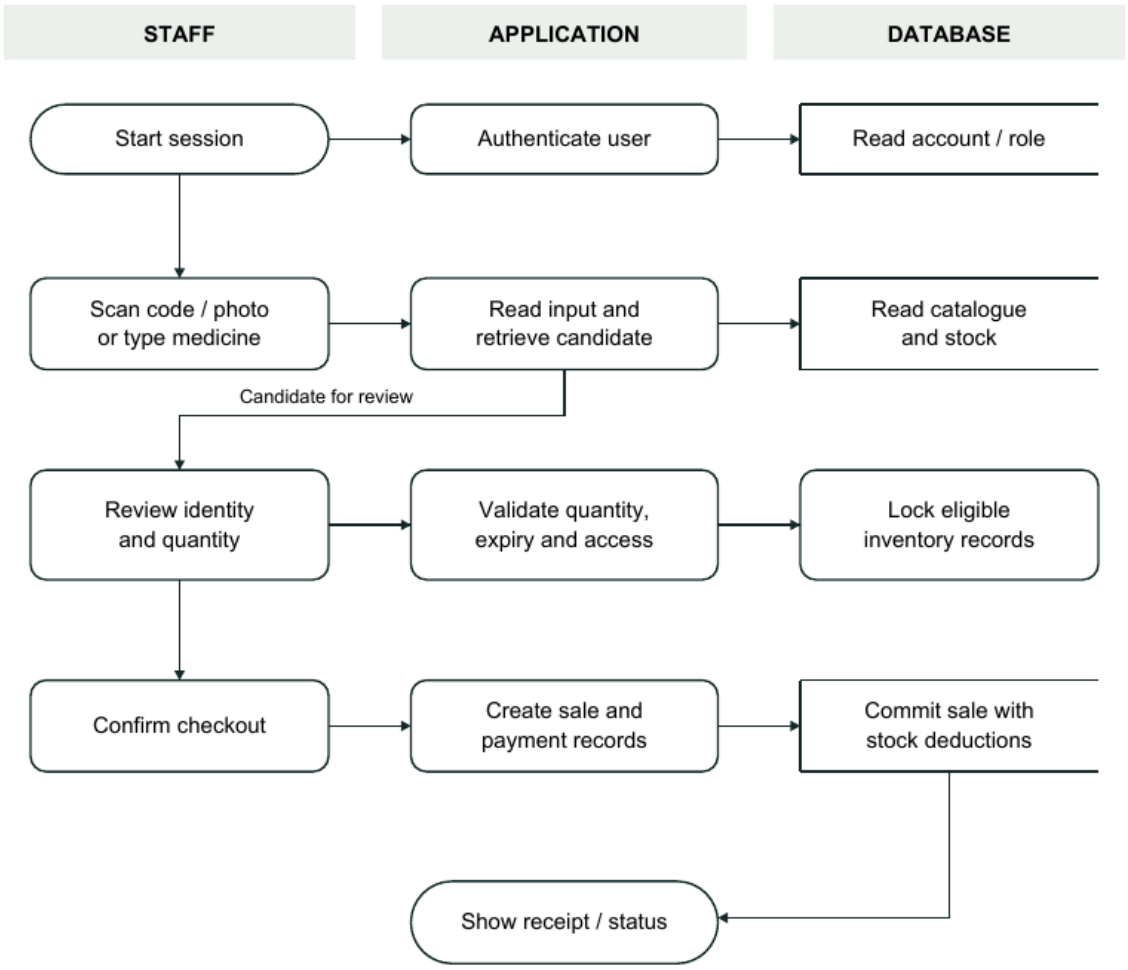


Figure 3 Activity Diagram

DFD 0 Context Diagram

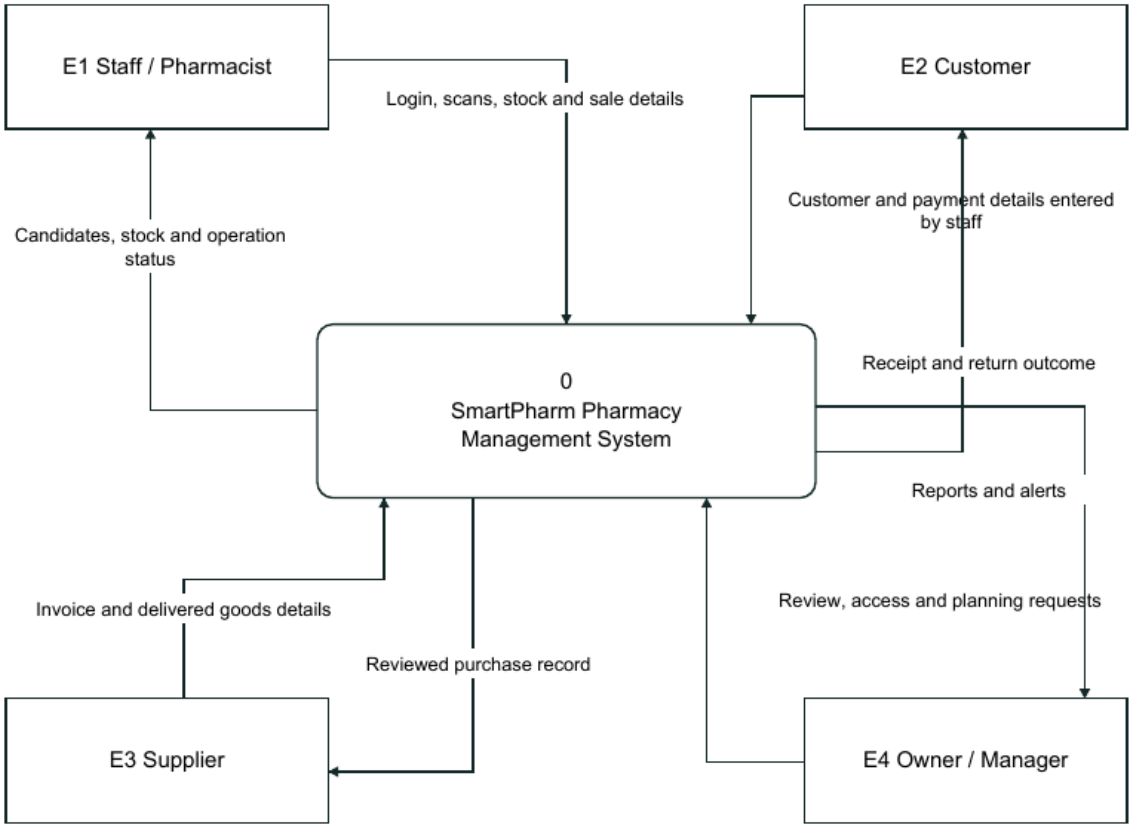


Figure 4 DFD 0 Context Diagram

DFD 1 Major Processes

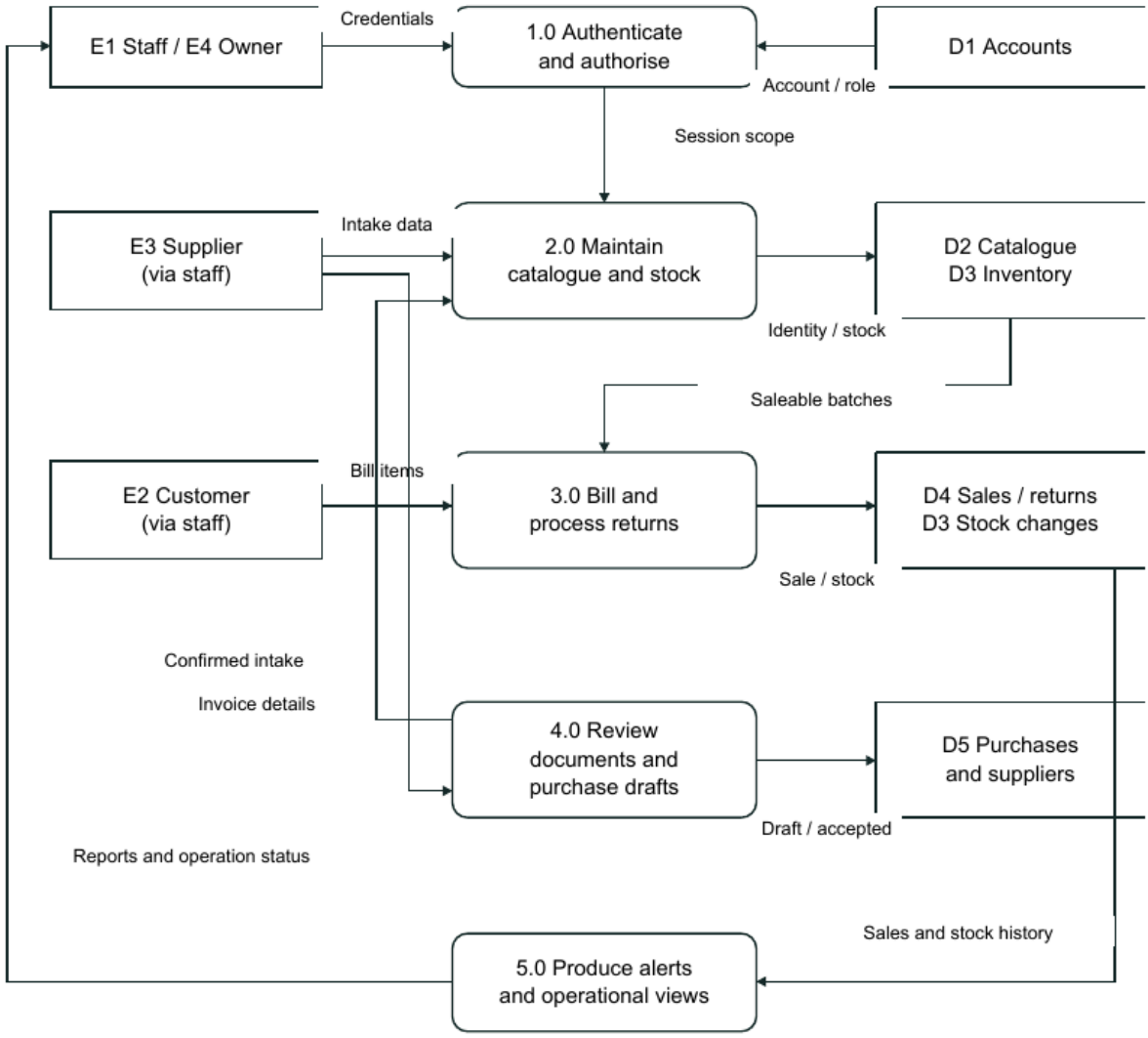


Figure 5 DFD 1 Major Processes

DFD 2 Billing Decomposition

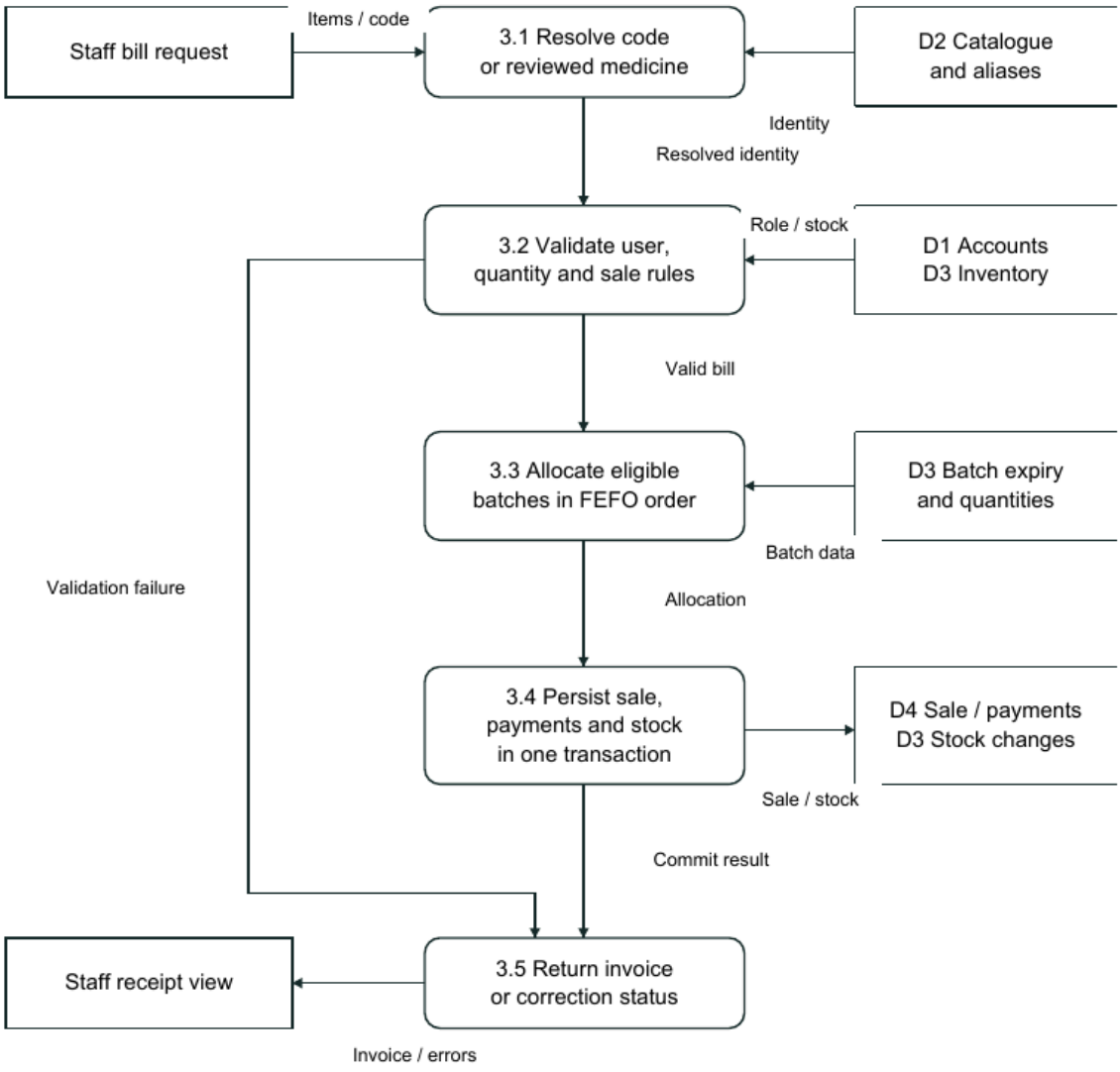


Figure 6 DFD 2 Billing Decomposition

System Flow

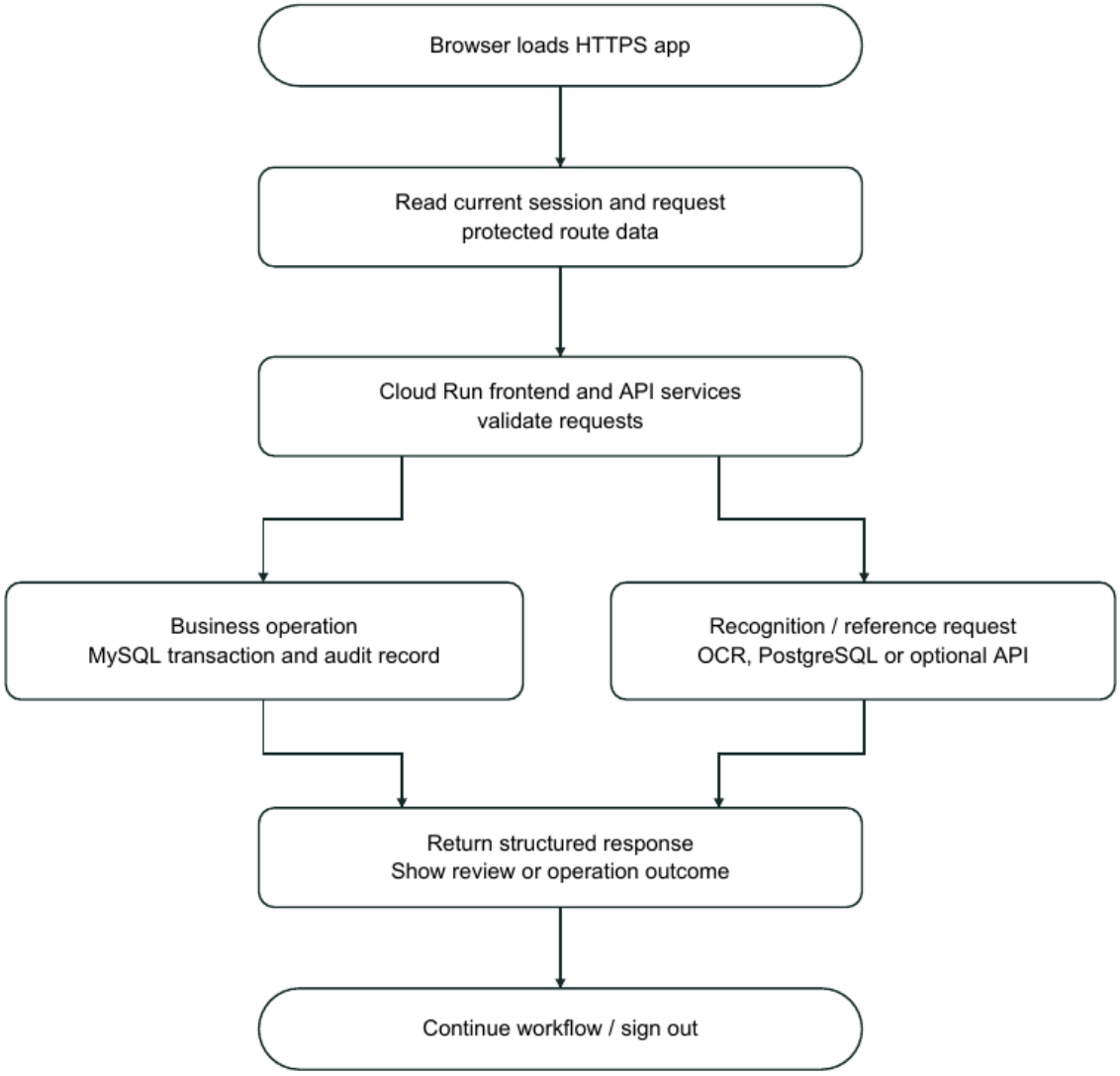


Figure 7 System Flow

Functional Flow

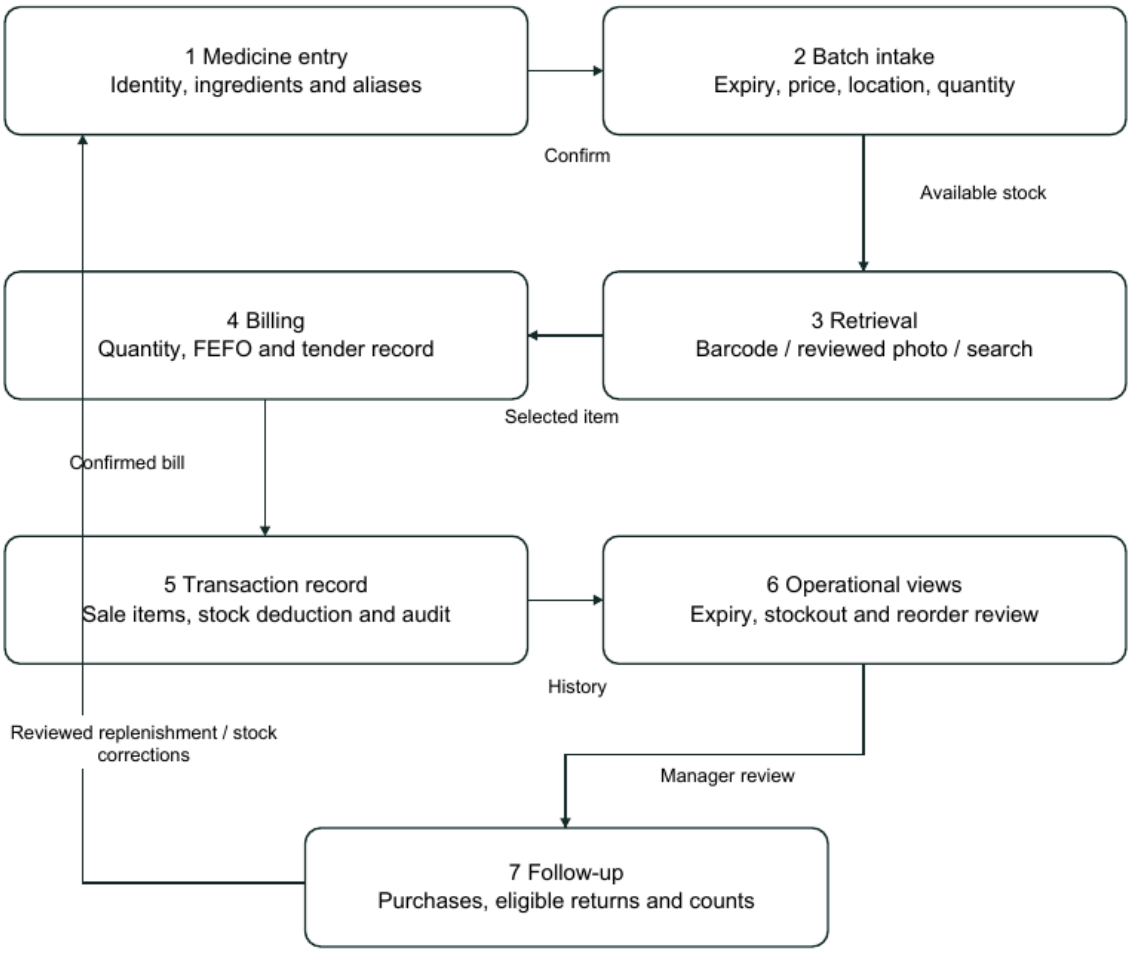


Figure 8 Functional Flow

Core Data Model

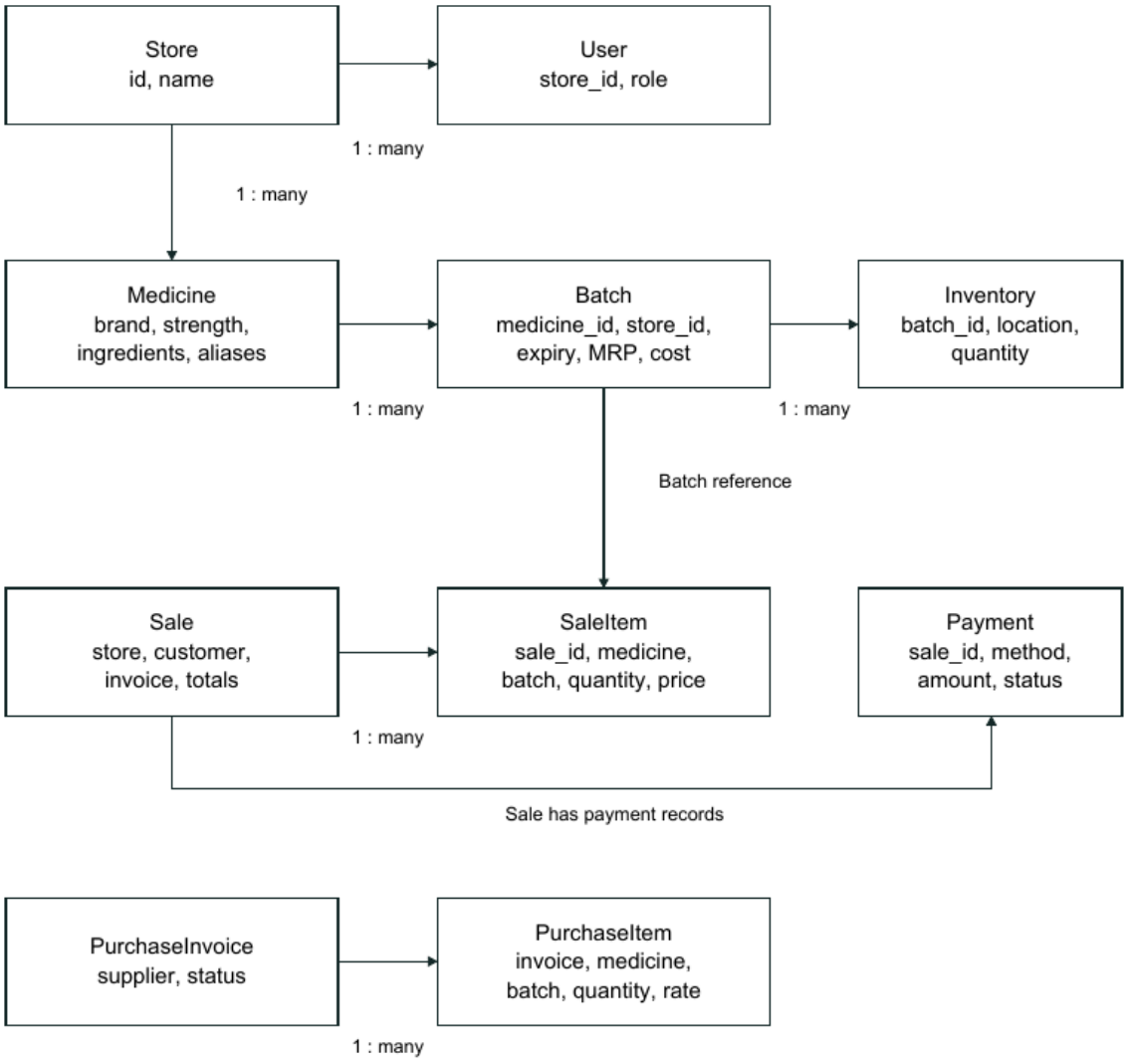


Figure 9 Core Data Model

Test Case Verification Diagram

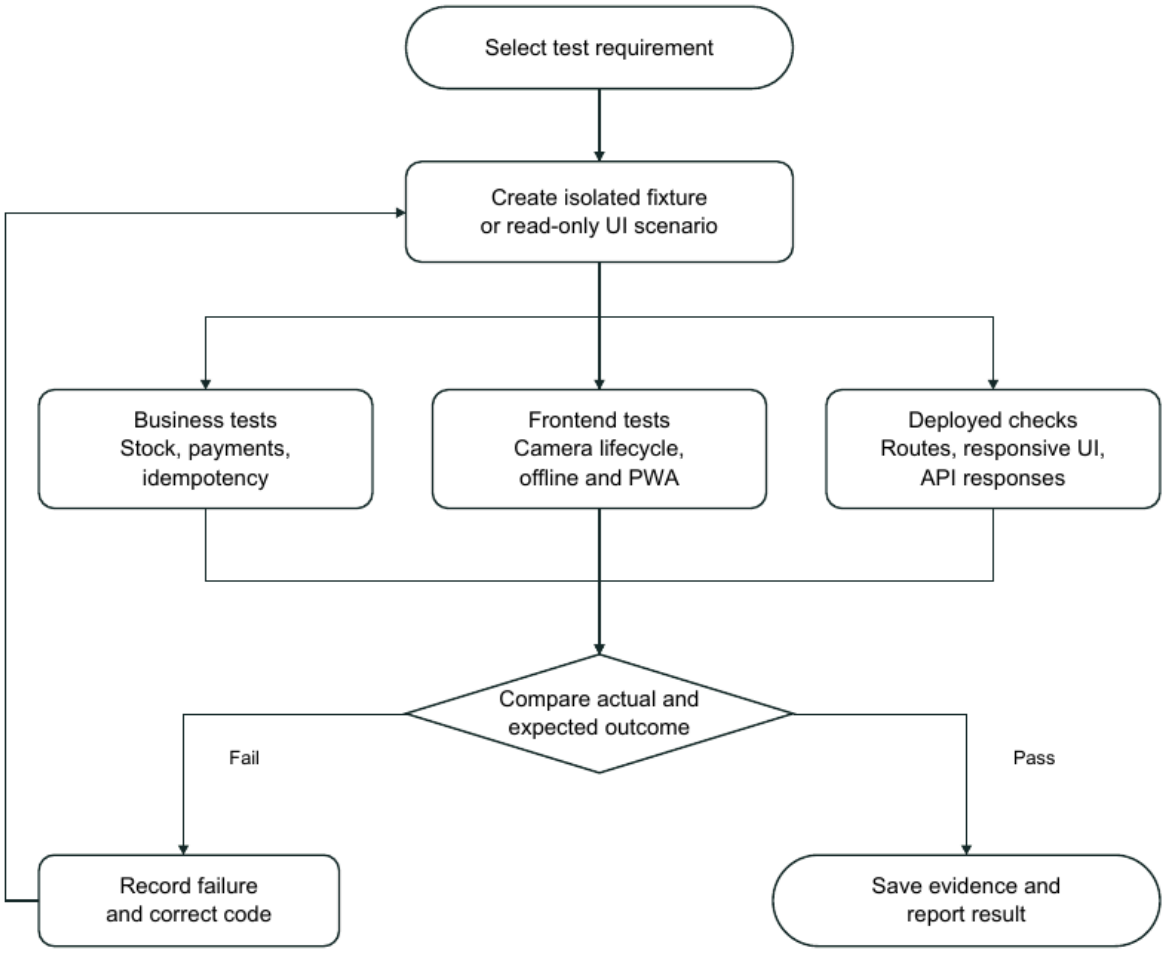


Figure 10 Test Case Verification Diagram

- API and integration inventory

Table 3.1 Business API inventory

Area	Methods and route examples	Purpose
Authentication	POST /api/auth/login; GET /api/auth/me; POST /api/auth/users	Sign in, retrieve account and manage staff.
Medicine catalogue	GET/POST /api/medicines; GET /api/medicines/search	Search and maintain store medicine identities.
Barcode	GET /api/medicines/barcode; POST /api/medicines/barcode-image	Resolve a mapped barcode or decode an image.
Inventory	GET/POST /api/batches; POST /api/inventory/adjust	Record batches and audited quantity adjustments.
Expiry / location	GET /api/inventory/expiry; GET /api/locations; GET /api/shelf/layout	Review expiry and physical storage.
Sales	GET/POST /api/sales; GET /api/sales/{id}	Create bills and retrieve transaction details.
Returns	GET /api/returns/eligible-items/{reference};	Check return eligibility and record

	POST /api/returns	reviewed returns.
Purchases	GET/POST /api/purchases; POST /api/purchases/{id}/accept	Maintain drafts and accept verified intake.
Suppliers / expenses	GET/POST /api/suppliers; GET/POST /api/expenses	Maintain supplier and operating-expense records.
Branches / audit	GET /api/multibranch/stores; GET /api/audit-logs	Inspect branches and operation history.

Table 3.2 Recognition and support APIs

API group	Route examples	Function
Vision	POST /api/vision/recognize; POST /api/vision/verify	Produce and verify packaging candidates.
Stocking	POST /api/vision/stocking/read, /search, /confirm	Read, search and confirm reviewed entry.
Distributor invoice	POST /api/distributor/scan-invoice; /verify-delivery; /confirm-delivery	Extract invoice rows and verify delivery.
Prescription	POST /api/ai/prescription/scan; POST /api/prescriptions/{id}/verify	Draft transcription and pharmacist review.
Planning / copilot	GET /api/ml/reorder-recommendations; POST /api/ai/copilot/query	Read estimates and query grounded records.
Offline sync	GET /api/sync/offline-pack; POST /api/sync/replay	Account-scoped catalogue and idempotent replay.

External integrations: Google Gemini for optional handwriting/copilot synthesis; Tavily for optional medicine-reference search; Aiven MySQL and PostgreSQL for server-side persistence; Google Cloud Run, Cloud Build, Artifact Registry and Secret Manager for hosting and configuration. Browser integrations include ZXing, camera permission, optional speech recognition and PWA APIs. No payment-gateway settlement or automatic supplier ordering is claimed.

- Working methodology and test cases

Verification combines isolated transactional tests, frontend tests, read-only live-service checks and browser interaction checks. The focused demo/ML group passed 7 tests and the frontend suite passed 28 tests. The frontend production build/type check passed; lint recorded 0 errors and 81 existing warnings. These are scoped recorded checks, not exhaustive or clinical validation.

ID	Input or action	Expected result	Evidence status
TC01	Import the eight-medicine demo fixture twice.	Second run creates no duplicate sales or stock.	Automated regression passed.
TC02	Compute imported stock and payment totals.	Stock is 480 - 217 + 20 = 283; line/payment totals balance.	Automated fixture passed.
TC03	Forecast a short series of selling days.	Use baseline; avoid recursive thousands-of-units suggestion.	Automated regression passed.
TC04	Open dashboard with stored demo activity.	Render chart, stock/expiry alerts and purchase drafts.	Live browser observed.
TC05	Open return lookup at 375 and 1440 px.	Icon does not overlap text; no page-level overflow.	Live dimensions checked.
TC06	Present mapped barcode frames; repeat same code.	Resolve code, suppress duplicate add and stop camera stream on close.	Simulated browser/component checks;

			physical device retest needed.
TC07	Scan a weak or rotated package photo.	Show review candidate or explicit recapture/manual fallback.	Workflow observed; general accuracy not quantified.
TC08	Load protected deep link offline.	Public reconnect screen; do not cache private response.	Prior PWA browser evidence.
TC09	Transcribe handwritten prescription.	Editable rows, warnings for ambiguity, no invented quantity.	Sample workflow observed; pharmacist review required.

Demonstration data: eight active medicines with sixteen front/back photographs; fifty added synthetic sales, twelve fictional customers, three fictional suppliers, five purchases including three drafts, six expenses and three new near-expiry batches. The isolated seed fixture balances 480 - 217 + 20 = 283 units; later live transactions can change stock.

- Frontend screenshots

Sign In

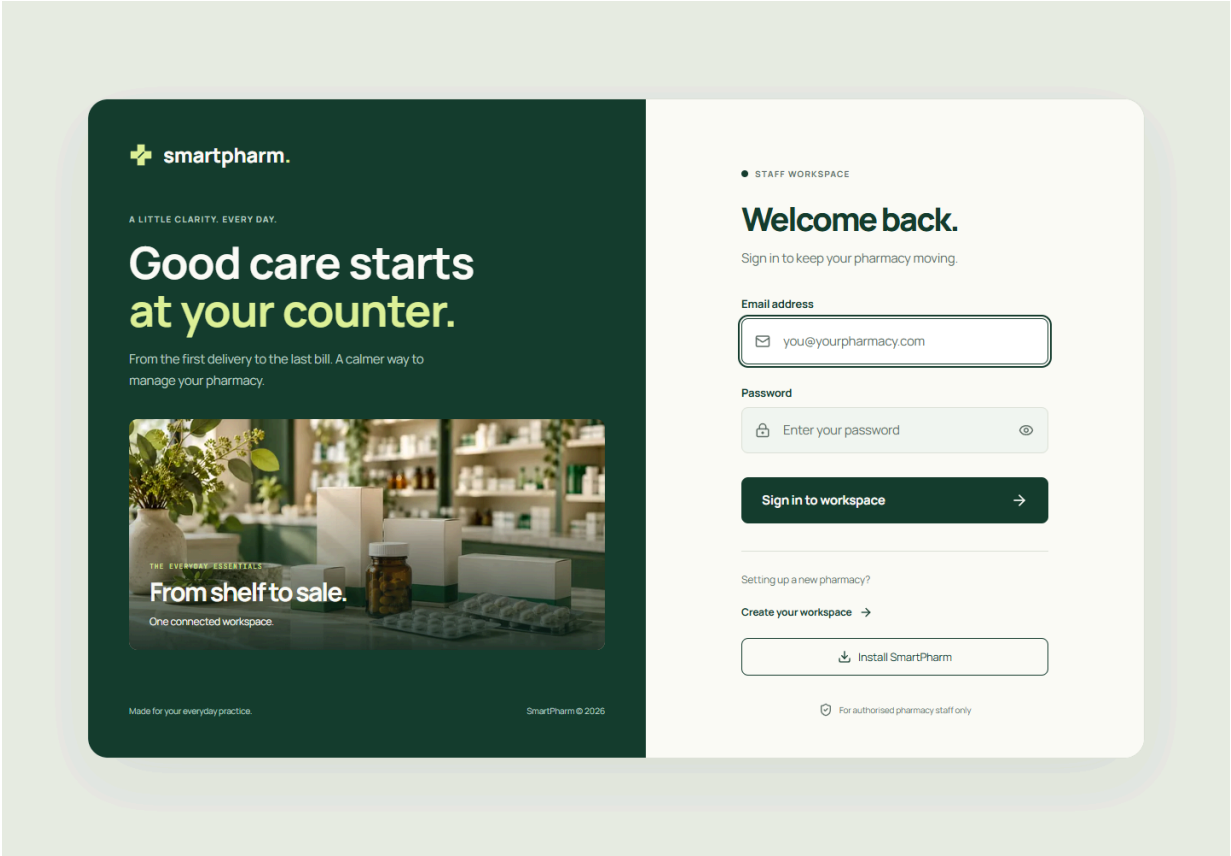


Figure 11 Sign In Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.

Dashboard

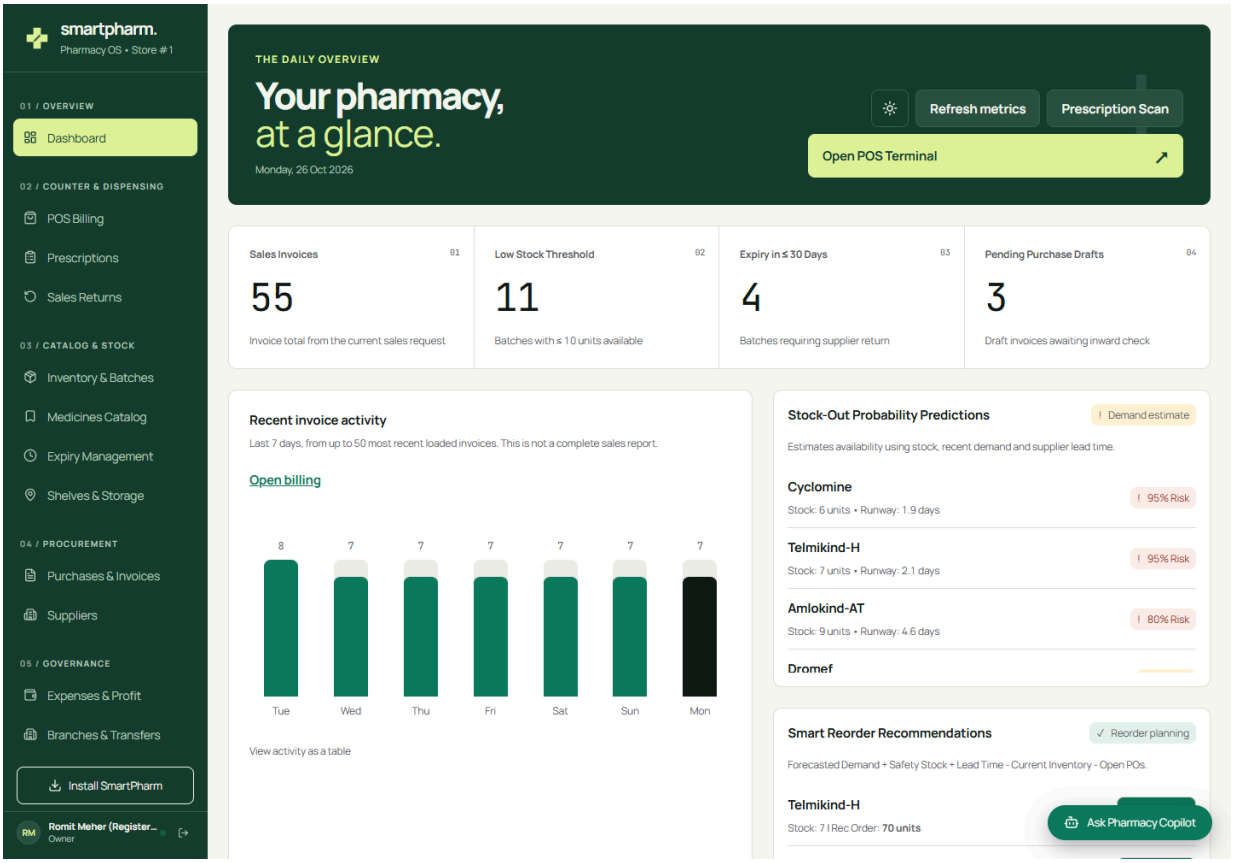


Figure 12 Dashboard Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.

POS Billing Terminal

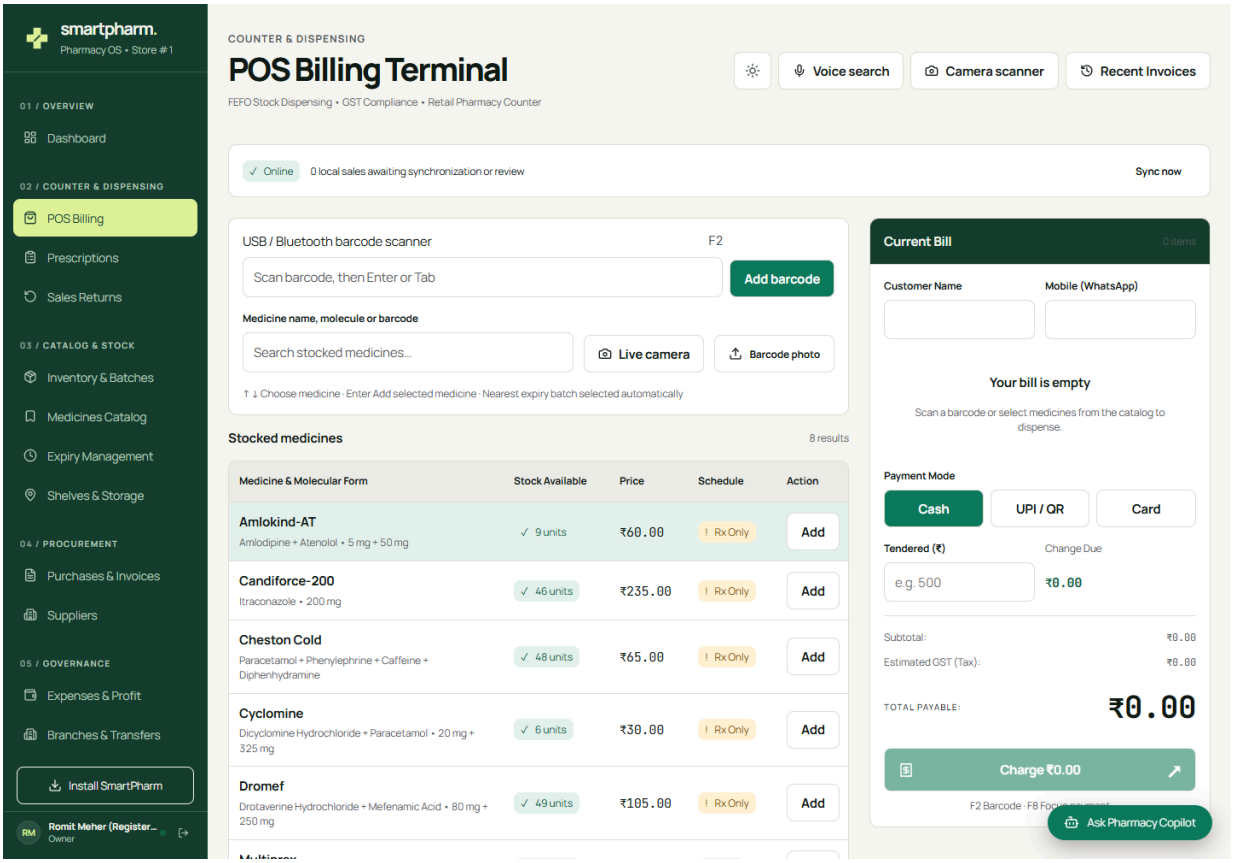


Figure 13 POS Billing Terminal Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.

Inventory and Batches

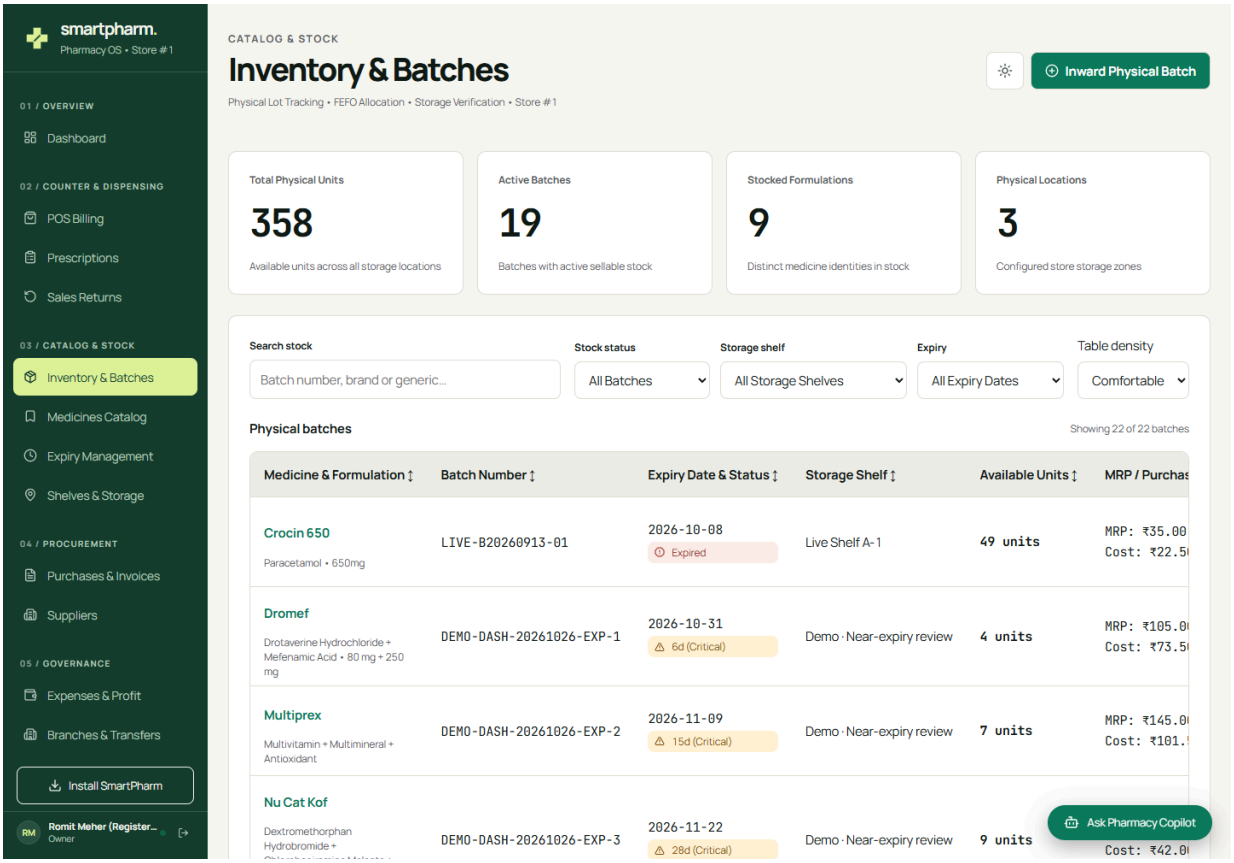


Figure 14 Inventory and Batches Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.

Medicines Catalogue

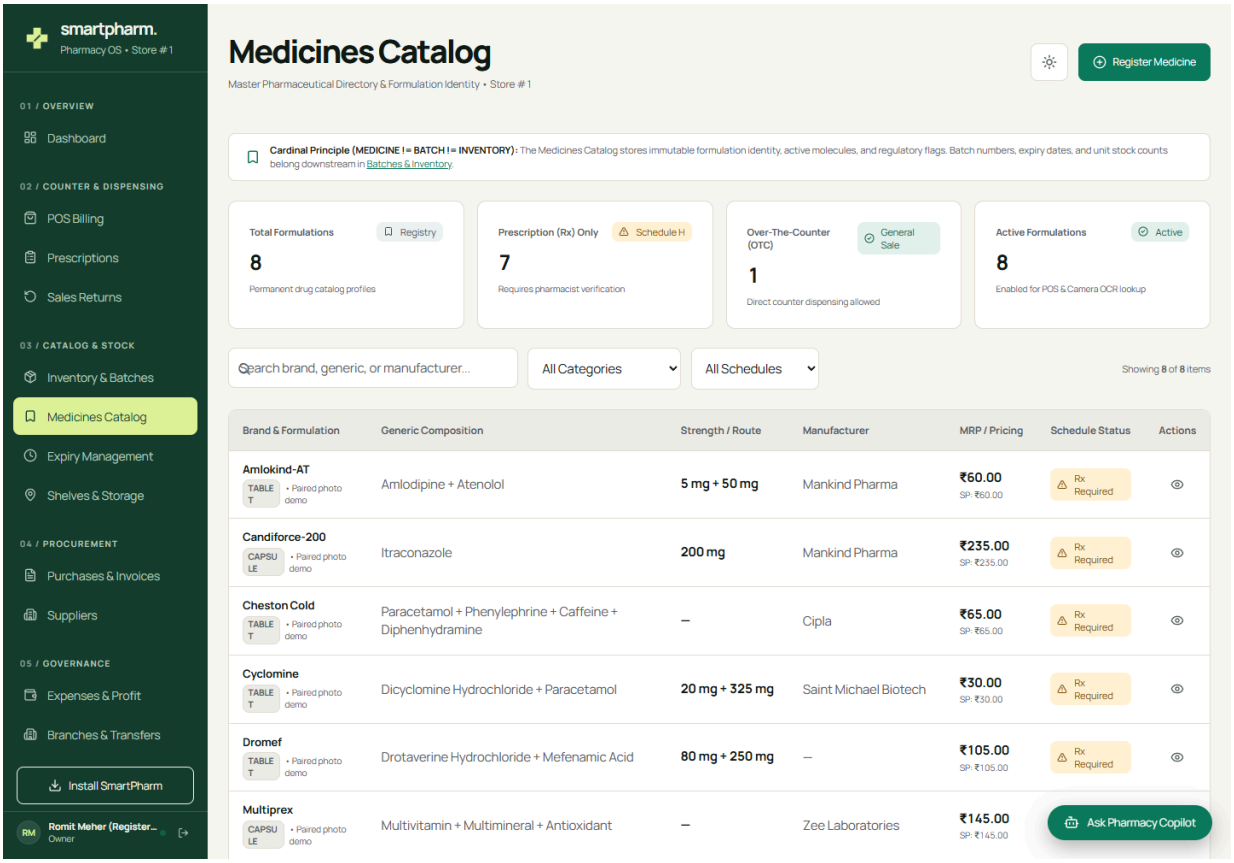


Figure 15 Medicines Catalogue Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.

Purchases and Invoices

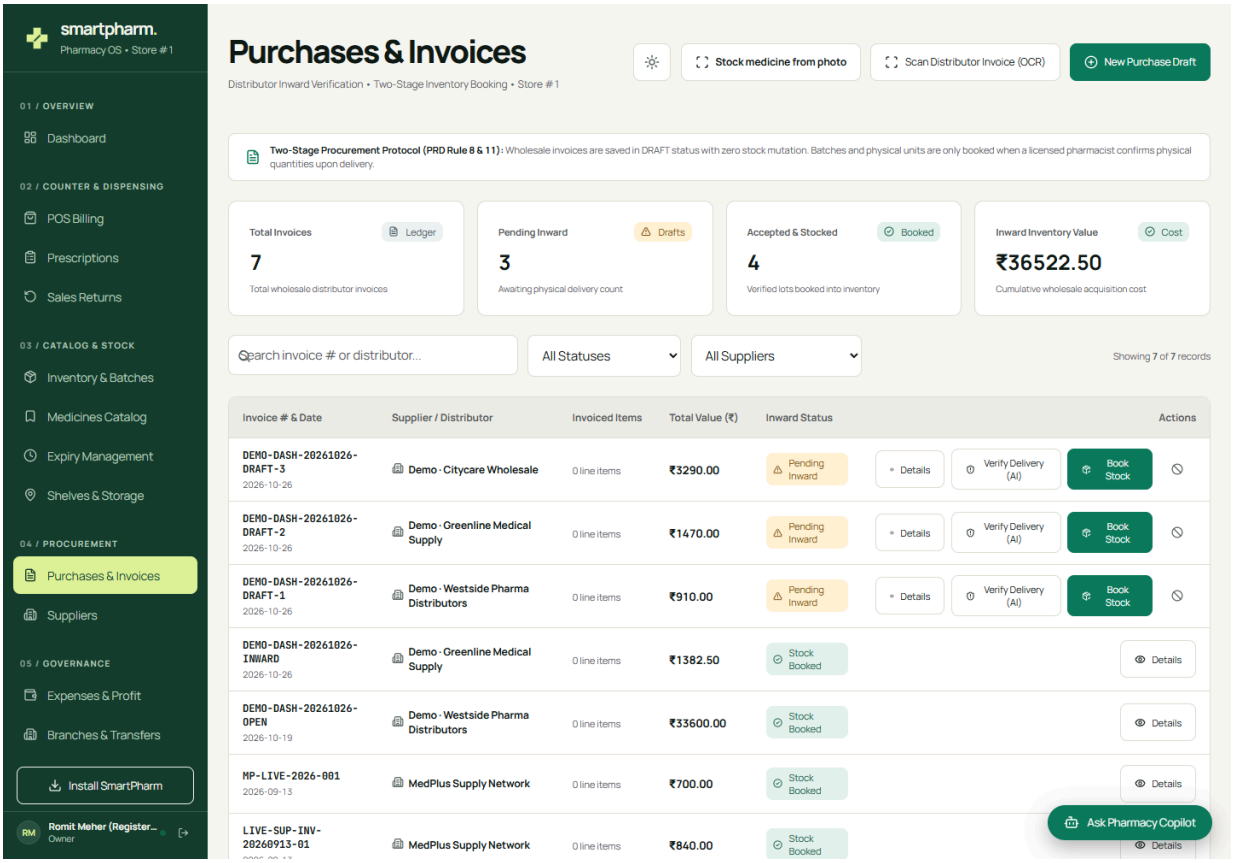


Figure 16 Purchases and Invoices Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.

Prescription Workspace

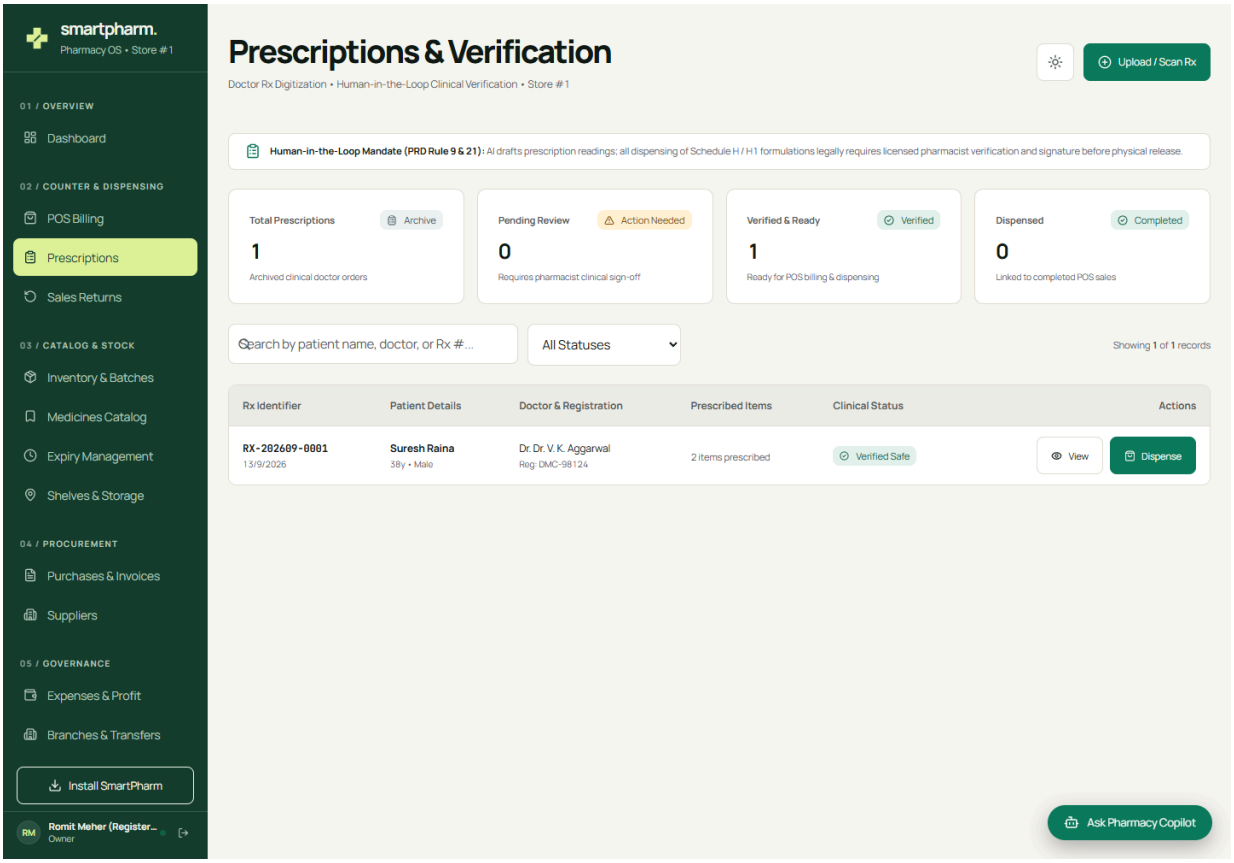


Figure 17 Prescription Workspace Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.

Expiry Management

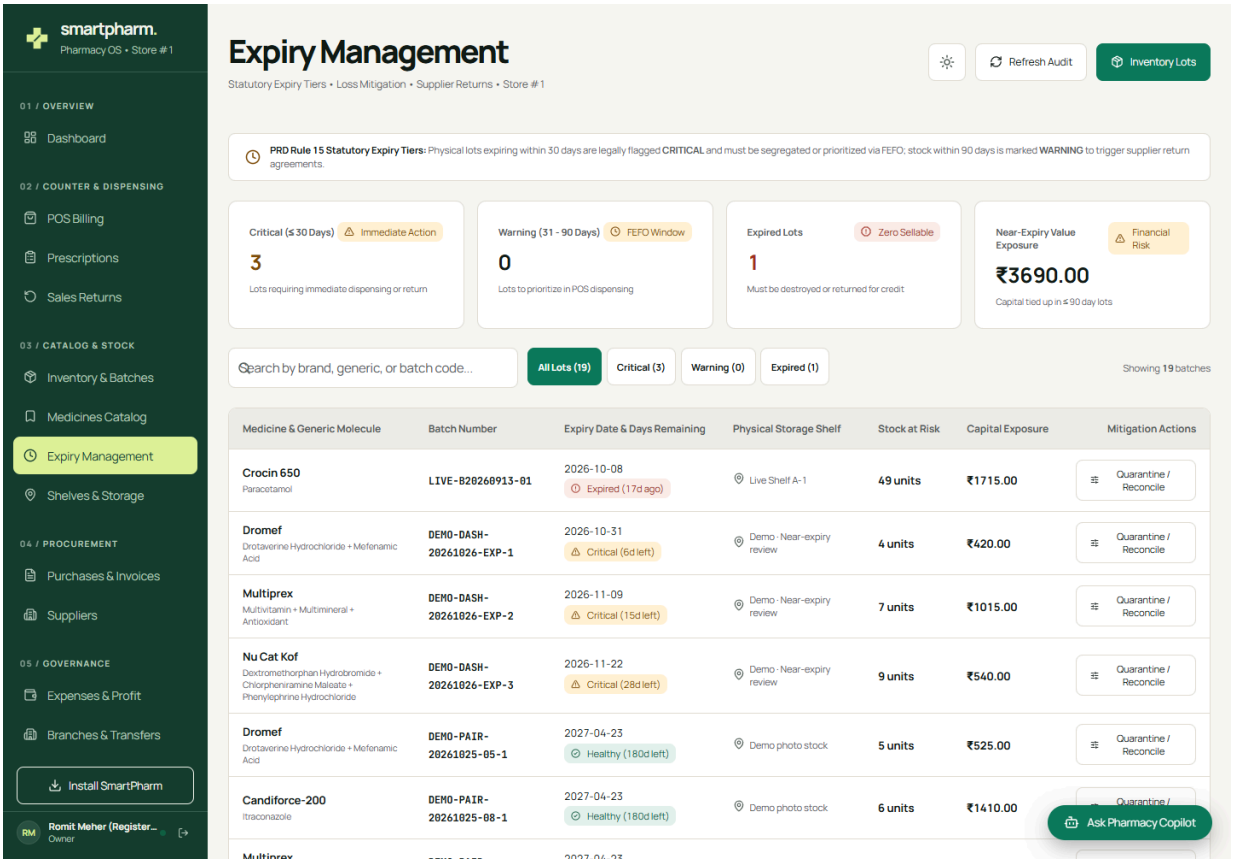


Figure 18 Expiry Management Actual Frontend

Actual deployed frontend captured on 26 October 2026. The screenshot documents visible functionality with classroom demonstration records; it does not submit a payment or stock operation.